

Mastery equations

1. Write the equation for acid and metal oxides. [1 mark]
2. Write the equation for acids and carbonates. [1 mark]
3. Write the equation for acids and metals. [1 mark]
4. Write the equation for acids and bases. [1 mark]

Mastery equations

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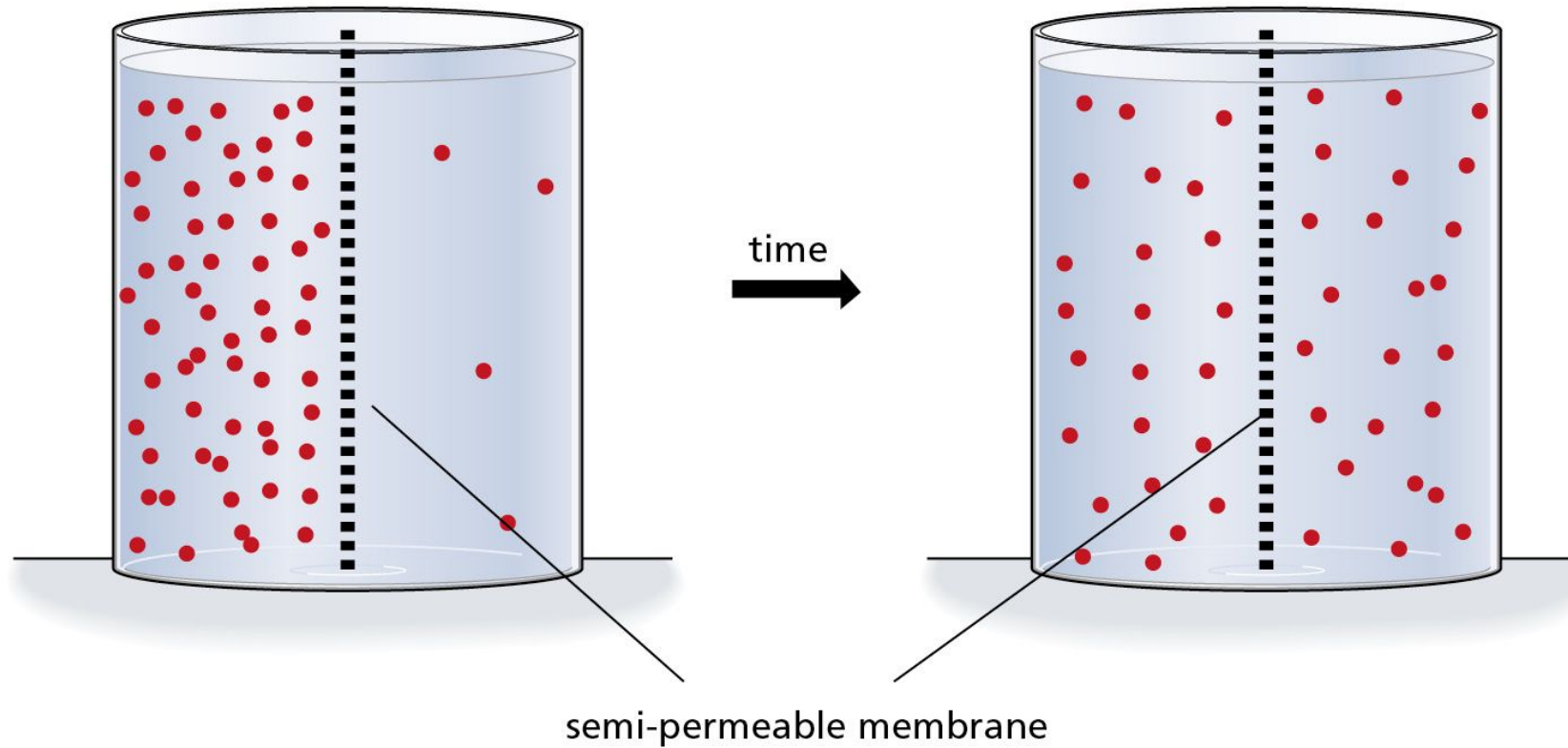


3. Write the equation for acids and bases. [1 mark]



Title – Gas Exchange

In your books describe what the image is showing. Add the drawing into your books.



Gas Exchange

Lesson 16



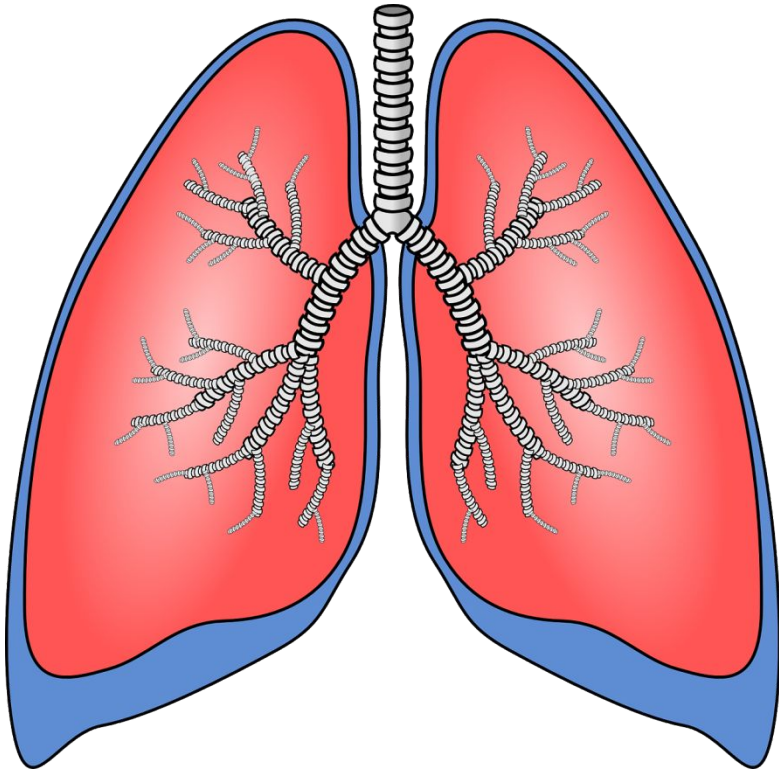
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Learning objectives:

By the end of the lesson you will be able to:

- Describe the mechanism of breathing in and out.
- Evaluate a model of breathing.
- Calculate changes in pressure and explain how these help us breathe.

Key Words
lungs
diaphragm
pressure



Did you know: The lungs of an average adult can hold up to 4.6 litres of air. There are three main gases in the air. Oxygen, carbon dioxide and nitrogen

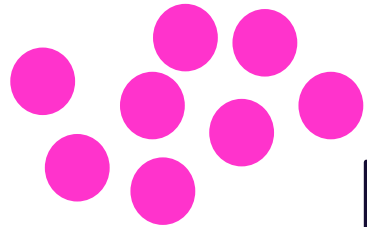
How do substances move
from the outside of a cell
to the inside of a cell?



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Diffusion

- Diffusion is the movement of particles from a high concentration to a lower concentration area.
- When will diffusion stop?
- This usually results in an equilibrium (the same number of particles on each side).
- What could increase the rate of diffusion (speed up the process)



ACTIVATION

The respiratory system are the organs and tissues that help your body to exchange gases between the air and the blood

OXYGEN gas
Goes in to the
lungs

CARBON
DIOXIDE gas
Comes out



Write this
information
down

Gas Exchange

Exchanging means to replace one thing with another.

OXYGEN (O_2) is exchanged with the CARBON DIOXIDE (CO_2) because carbon dioxide is a waste product and need to be



What is the difference between breathing and respiration?

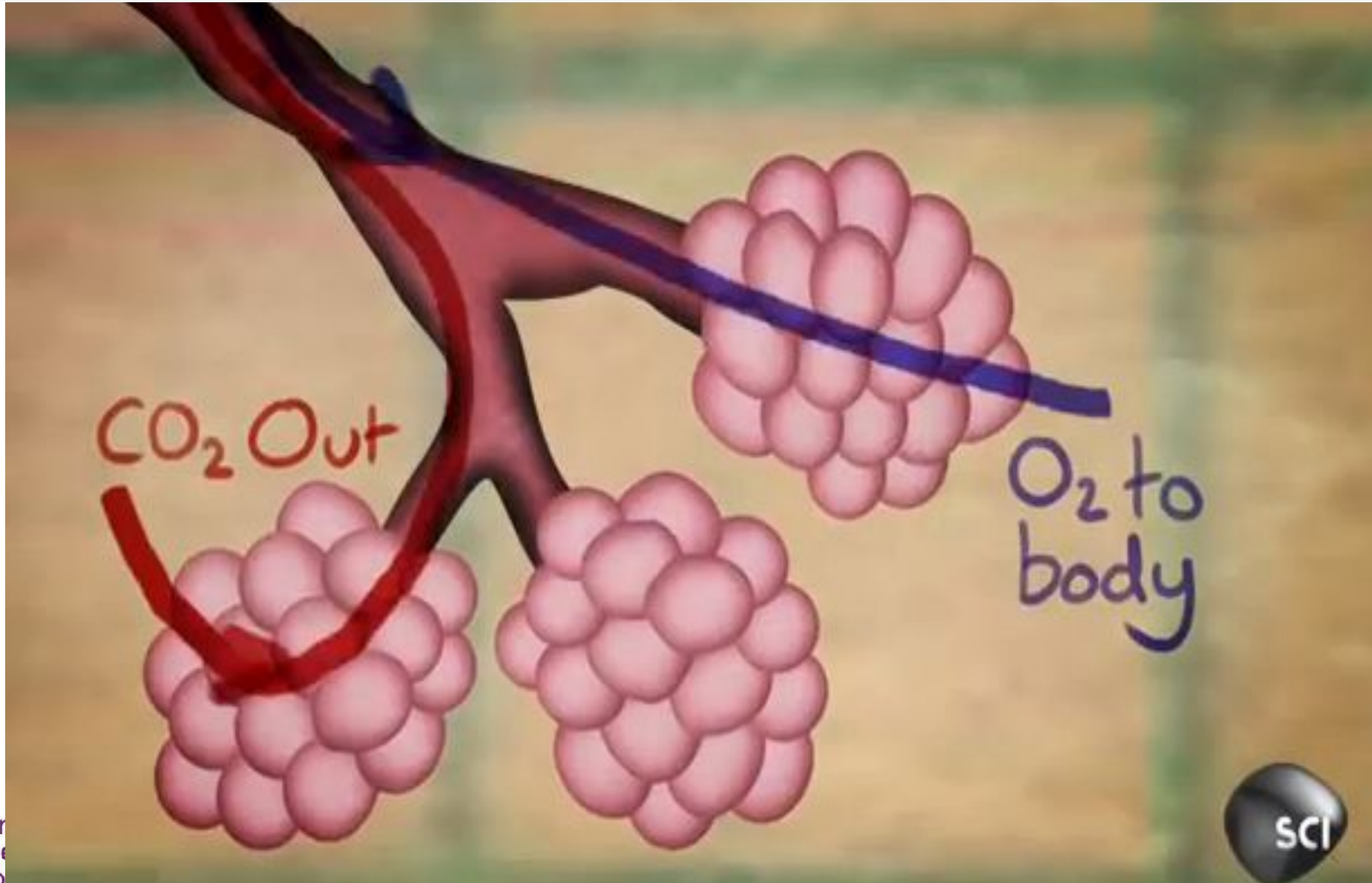
Breathing is the movement of O_2 and CO_2 into and out of the lungs (TAKING IN)

Respiration is the use of oxygen to make energy (MAKING ENERGY)

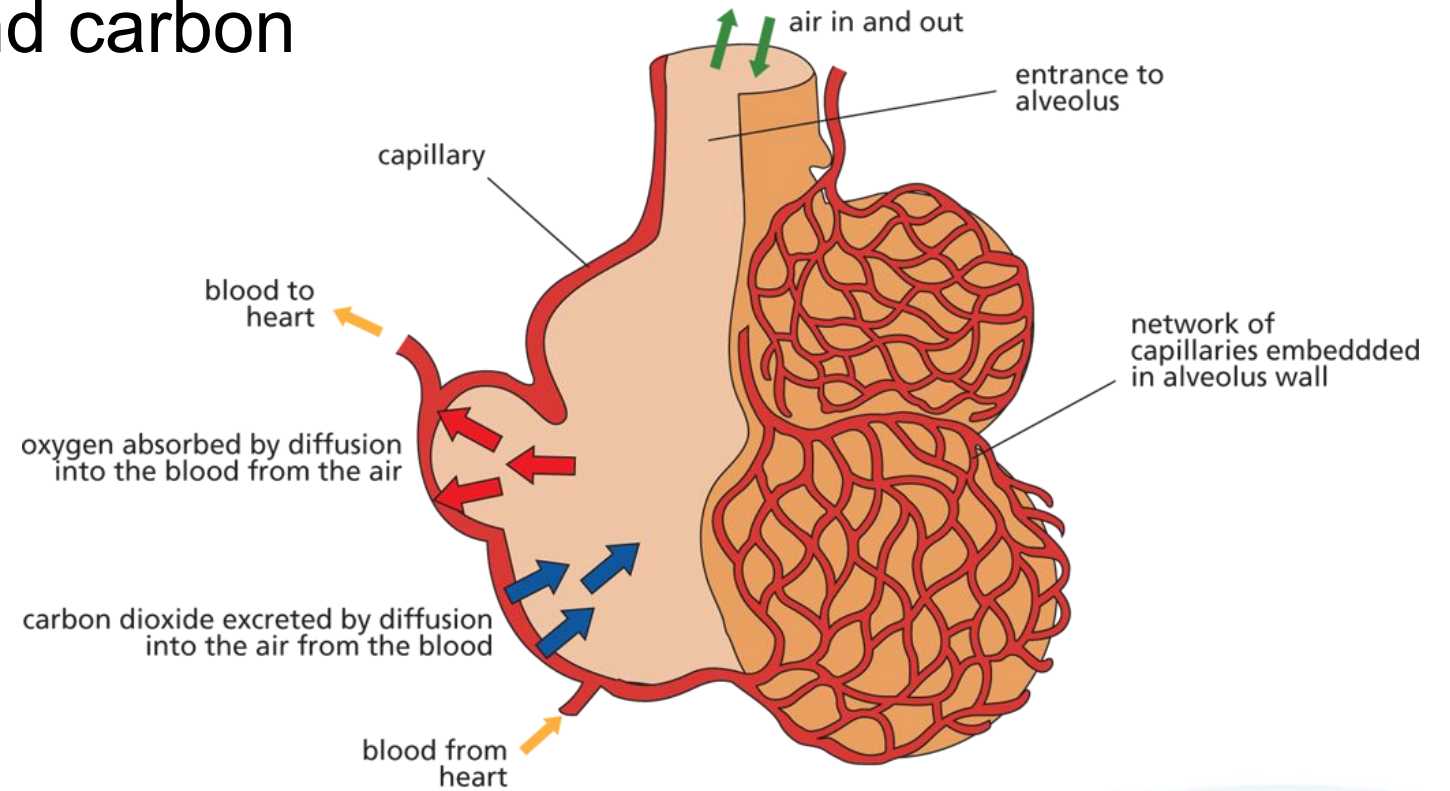
Could you list parts of the body to do with breathing

NOSE
MOUTH
THROAT
LUNGS
TRACHEA

The Alveoli where gas exchange happens

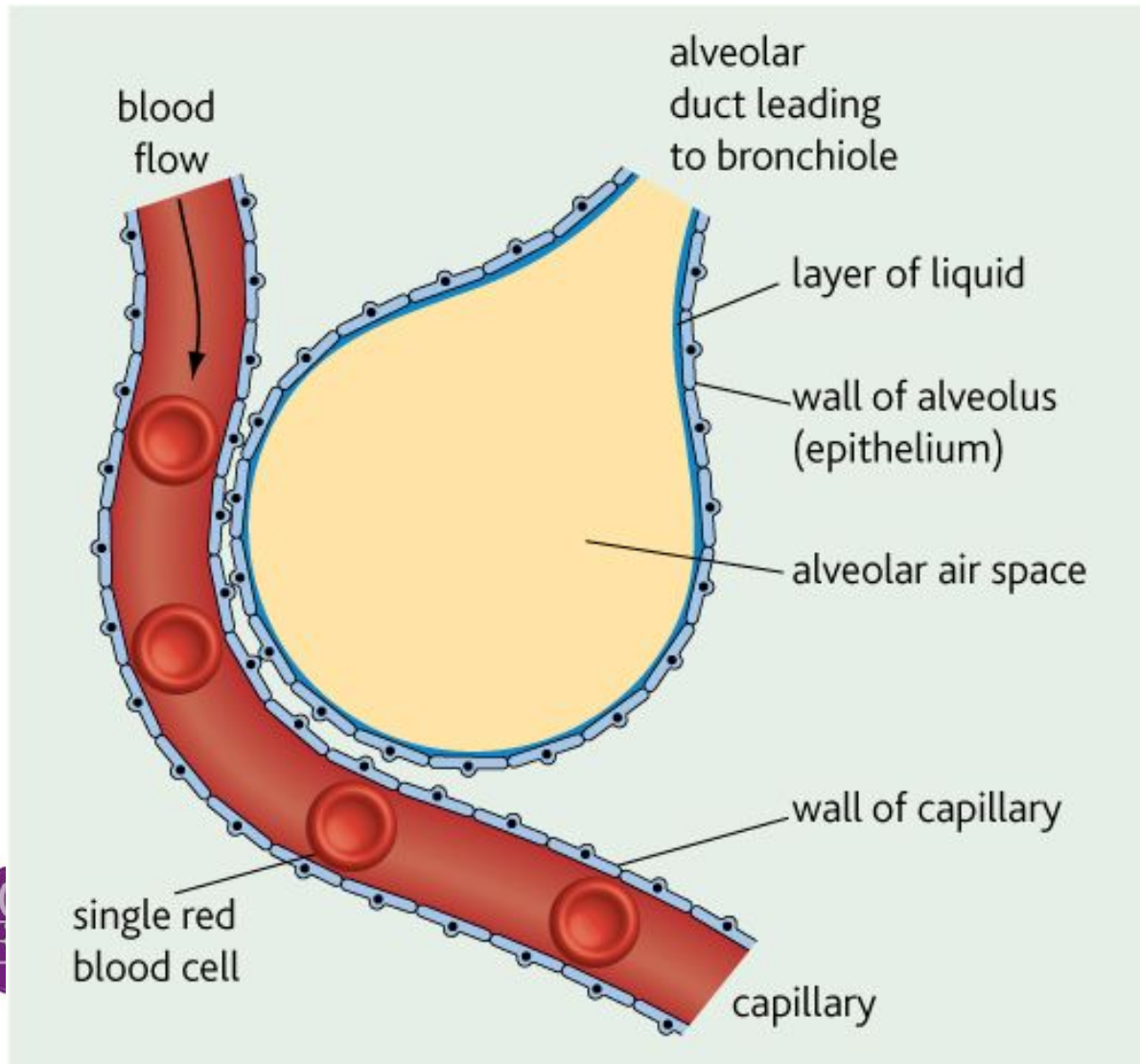


These gases move across the alveoli and capillary walls by diffusion. Diffusion is the spreading out of particles. In the lungs, the particles are oxygen and carbon dioxide.



ACTIVATION

How are the alveoli adapted for efficient gas exchange? (3 marks)



Many alveoli, so large surface area for rapid diffusion, rich blood supply to maintain diffusion gradient, small distance between air and blood, so less distance for diffusion, moist surface for gases to dissolve

CONSOLIDATION

Respiration or breathing?

- *What is the difference between respiration and breathing?*
- *Breathing- process of gases getting in and out of the body*
- *Respiration- process in living things in which oxygen is used to release the energy from food.*



Activity: Complete this paragraph fill **Choose 1 to complete**

The air first enters through the N____. It then travels down the T_____ into the L____. Gas exchange is when ox_____ gas is replaced by ca_____ di_____ in the lungs

When we inhale _____ gas goes into the lungs. When we exhale _____ gas comes out of the lungs. We call this g_____ e_____

When we inhale _____ gas goes into the lungs. When we exhale _____ gas comes out of the lungs. We call this _____. The _____ is the last part of the journey for air.

Review

The air first enters through the **NOSE**. It then travels down the **TRACHEA** into the **LUNGS**. Gas exchange is when **OXYGEN** gas is replaced by **CARBON DIOXIDE** in the lungs

When we inhale **OXYGEN** gas goes into the lungs. When we exhale **CARBON DIOXIDE** gas comes out of the lungs. We call this **GAS EXCHANGE**

When we inhale **OXYGEN** gas goes into the lungs. When we exhale **CARBON DIOXIDE** gas comes out of the lungs. We call this **GAS EXCHANGE**. The **ALVEOLI** is the last part of the journey for air.

Task- Answer worksheet